

🏥 Healthcare Appointment & Resource Management System (ICEPOT-HPRM)

1. Problem Statement

Hospitals and clinics often face **appointment scheduling conflicts** and **resource bottlenecks** (e.g., doctors overbooked, diagnostic equipment unavailable, overlapping patient slots).
This leads to **delays in patient care**, **inefficient resource utilization**, and **poor patient experience**.

🎯 Objective

- To design a **Healthcare Patient Appointment & Resource Management System** that:
- Ingests requests from **emails** and **ADO work items**
 - Manages patient appointments, doctor schedules, and equipment availability
 - Tracks SLA compliance for patient wait times
 - Provides dashboards for administrators to optimize resources

2. Input Sources

Source	Type	Description
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Email	Unstructured	Patients or staff send appointment or reschedule requests
ADO Work Item	Structured	Work items created for scheduling tasks or resource allocation
Web Portal	Structured	Direct patient appointment booking
EHR Integration	Semi-structured	Sync with patient records and doctor availability

3. System Architecture Overview

flowchart TB

PATIENT[Patient/Staff] --> | Email or ADO Work Item | INGEST[Ingestion Service]

INGEST --> PARSE[Email & ADO Parser]

PARSE --> CLASSIFY[AI Classification Service]

CLASSIFY --> DB[MongoDB - Appointment Repository]

DB --> QUEUE[SQS/SNS - Event Queue]

QUEUE --> ASSIGN[Scheduling Service]

ASSIGN --> NOTIFY[Notification Service]

ASSIGN --> SLA[SLA Tracking Service]

NOTIFY --> | Email/SMS | PATIENT

NOTIFY --> | In-app | DOCTOR[Doctor]

DOCTOR --> UPDATE[Update Status API]

UPDATE --> DASH[Dashboard Service]

DASH --> ADMIN[Hospital Administrator]

| Module | Description |

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| ****Ingestion Service**** | Captures appointment requests from email and ADO APIs |

| ****Parser**** | Extracts patient details, doctor ID, preferred time |

| ****AI Classification**** | Categorizes request (new appointment, reschedule, cancellation) |

| ****Scheduling Service**** | Allocates doctor and equipment based on availability |

| ****SLA Tracking**** | Monitors patient wait times and appointment delays |

| ****Notification Service**** | Sends confirmations via email/SMS |

| ****Dashboard Service**** | Displays real-time appointment and resource utilization |

| ****Audit & History**** | Maintains full lifecycle logs for compliance |

[*] --> New: Patient submits request

New --> Parsed: System extracts metadata

Parsed --> Classified: AI assigns category and priority

Classified --> Scheduled: Doctor and equipment allocated

Scheduled --> Confirmed: Patient notified

Confirmed --> InProgress: Appointment ongoing

InProgress --> Completed: Appointment finished

Completed --> Closed: Records updated in EHR

Closed --> [*]

| Method | Path | Purpose |

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| POST | /api/v1/appointments/email | Ingest appointment request from email |

| POST | /api/v1/appointments/ado | Ingest appointment request from ADO work item |

| GET | /api/v1/appointments | List all appointments (filterable by doctor, patient, status, SLA) |

| GET | /api/v1/appointments/{id} | Get details of a specific appointment |

| PATCH | /api/v1/appointments/{id} | Update appointment fields (status, doctor, time) |

| POST | /api/v1/appointments/{id}/assign | Assign doctor/equipment manually |

| POST | /api/v1/appointments/{id}/comments | Add comment or update |

| POST | /api/v1/appointments/{id}/attachments | Upload medical documents |

| POST | /api/v1/appointments/{id}/confirm | Confirm appointment |

| POST | /api/v1/appointments/{id}/reschedule | Reschedule appointment |

| POST | /api/v1/appointments/{id}/cancel | Cancel appointment |

| GET | /api/v1/appointments/{id}/history | Retrieve audit trail |

| GET | /api/v1/dashboard/summary | Summary of scheduled/completed appointments |

| GET | /api/v1/dashboard/resources | Doctor and equipment utilization |

| GET | /api/v1/dashboard/sla | SLA breach and at-risk appointments |

| GET | /api/v1/reports | Generate analytics reports |

| GET | /api/v1/notifications | Fetch patient/doctor notifications |

| PATCH | /api/v1/notifications/{id}/read | Mark notification as read |

| GET | /api/v1/doctors | List doctors and availability |

| GET | /api/v1/equipment | List medical equipment availability |

| POST | /api/v1/escalations | Escalate critical scheduling conflicts |

| GET | /api/v1/compliance/audit | Retrieve compliance audit logs |

| Priority | Response SLA | Resolution SLA |

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Critical (Emergency)	15 min	Immediate scheduling
High (Urgent care)	1 hour	Same-day appointment
Medium (Routine checkup)	4 hours	Within 2 business days
Low (Follow-up)	1 business day	Within 5 business days

Role	Permissions
Patient	Submit, view, reschedule, cancel
Doctor	View assigned, confirm, update status
Nurse/Admin	Manage schedules, reassign, monitor SLA
Hospital Administrator	Manage users, departments, and analytics

AWS deployment architecture

flowchart TB

U[Users] --> CF[CloudFront]
CF --> S3[S3 - React static build]
U --> ALB[Application Load Balancer]
ALB --> ECS1[ECS Fargate - Appointment API]
ECS1 --> ECS2[ECS Fargate - Notification Service]
ECS1 --> ECS3[ECS Fargate - Dashboard Service]
ECS1 --> RDS[RDS PostgreSQL - Multi-AZ]
ECS1 --> REDIS[(ElastiCache Redis)]
ECS1 --> SQS1[SQS - Appointment Events]
SQS1 --> ECS4[ECS Fargate - Audit Service]